Kirolos Sedra

Software Engineer

Waterloo, Ontario | Relocatable | Authorized to work in Canada; no sponsorship required

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Professional Summary

Software Engineer with **3+ years of software-development experience** across **Java/Spring Boot fintech microservices**, full-stack web systems, and cloud-integrated IoT. Strong in REST APIs, SQL optimization, backend architecture, React/TypeScript, CI/CD, automated testing, debugging, and end-to-end system ownership from requirements through deployment and production support.

Experience

Aquasensing **Sep 2024 – Present**

Software Engineer *Part-time, Startup Environment*

- Developed full-stack features for a cloud-integrated **BLE–LTE IoT platform** using **React/TypeScript**, **Django/Python**, **Node.js**, Azure IoT Hub, Logic Apps, and Azure SQL-backed workflows; also supported firmware flashing and physical connectivity validation.
- Built an **Azure CLI** test-and-provisioning suite for a **70-unit IoT gateway fleet** and created MQTT alarm stubs to reproduce gateway events, isolate cloud-side defects, and reduce dependence on physical hardware during debugging.
- Designed an **ESP32-based automated end-to-end validation kit** that triggered sensor events and verified the complete alarm path through BLE/LTE connectivity, cloud processing, and email/SMS notifications; authored the SOP and trained **5 engineers** on its use.
- Led B2B technical discussions with a Canadian telecom/IoT provider and delivered a stakeholder demonstration that contributed to a **50-gateway commercial deployment** for an energy-sector customer, followed by continued technical support.

LINC One **Live** **GitHub** **May 2026 – Present**

Product Owner & Software Engineer *Volunteer*

- Led product ownership and engineering of a production **React/TypeScript/Firebase** platform serving the full **30+ member congregation**, coordinating three GitHub contributors and stakeholder feedback across scheduling, attendance, assessments, people development, administration, and ministry workflows.
- Evolved sensitive operations behind a **Hono/Cloudflare Worker REST backend** with Firebase ID-token authentication, server-side authorization, Zod validation, Brevo integrations, and **audited administrator self-service** for roles, forms, media/carousel configuration, and attendance, reducing recurring developer intervention.
- Directed AI-assisted engineering with **ChatGPT/Codex**, reviewing and debugging generated changes rather than treating them as trusted output; integrated **Bezalel AI** into selected product workflows and established backend CI/CD with tests, type checking, automated deployment, and production smoke checks.
- Owned a **three-phase product roadmap** spanning platform maturity, unified web consolidation, and future mobile/infrastructure expansion while prioritizing stakeholder needs, technical debt, security improvements, feature pruning, and release scope.

Ejada Systems **Sep 2023 – Aug 2024**

Software Engineer *Full-time, Hybrid*

- Developed and maintained **REST APIs** and backend **microservices** for fintech loan products using **Java Spring Boot**, **SQL**, and stored procedures, supporting systems for Al Rajhi Bank and Emkan Finance.
- Architected and developed a new **BNPL application** for Emkan Finance in a two-person core team, owning backend architecture, business logic, and system integration with the bank's existing frontend template.
- Improved measured SQL-query performance by **30%** through indexing and query optimization, and improved API performance by **50%** by eliminating redundant stored-procedure calls.

Selected Software Projects

EurekaVault **Live** **GitHub** *React 19, TypeScript, Firebase, Spring Boot*

- Owned the architecture and development of a live full-stack **prompt-management and AI workflow platform** integrating hierarchical prompt/version management, diffs and restore, attachments, directed relationships, reusable mindsets/preferences, semantic retrieval, prompt repurposing/mixing, and visual typed-DAG pipelines.
- Built the product with **React 19/TypeScript/Vite**, Firebase, Gemini-backed services, automated tests, and GitHub Actions; established a **Spring Boot 4.1 / Java 21** backend foundation on Render while leading an incremental migration from the current Firebase/Netlify architecture toward a three-tier REST architecture.

Laboratory Inventory and Experiment Management System *React, TypeScript, Firebase*

- Architected and developed a responsive inventory platform with complete **CRUD** workflows for equipment, quantities, serial numbers, locations, health status, lending, movement histories, projects, tags, comments, and photographs.
- Implemented per-serial tracking, experiment-feasibility validation, tag-based search, mobile workflows, and configurable **PDF reports** containing selected equipment, photographs, notes, health information, and optional movement logs.

Mobile Robot Wireless Performance Analysis *Python, Wi-Fi, 5G NSA/SA, iperf3*

- Designed and executed controlled outdoor, indoor, and RF-chamber experiments comparing Wi-Fi with 5G NSA/SA for mobile-robot connectivity, collecting latency, throughput, RSSI, MCS, and SINR measurements across repeated runs.
- Developed Python-based analysis workflows using motion-aligned windows and per-run P50/P99 statistics to isolate trajectory, load, and signal-condition effects and produce publication-quality figures and tables.

Education

University of Waterloo **Sep 2024 – Aug 2026**
Master of Applied Science in Electrical and Computer Engineering *92% (A+)*
Specialization: Pattern Analysis and Machine Intelligence

Ain Shams University **Sep 2018 – Jul 2023**
Bachelor of Science in Computer Engineering *CGPA: 3.12/4.00*

Technical Skills

Programming: Java, SQL, TypeScript, JavaScript, Python, C/C++, Bash, HTML, CSS
Backend & Web: Spring Boot, REST APIs, microservices, Django, Hono, React, Firebase, Swagger/OpenAPI, Maven
Databases & Data: Azure SQL, Firebase Realtime Database, stored procedures, indexing, query optimization
Testing & Quality: Spring Boot testing, Vitest, API testing, SIT, regression testing, production smoke testing, mocked external-service testing
Cloud & DevOps: GitHub Actions, Docker, Azure, Azure CLI, Azure IoT, Azure DevOps, Cloudflare Workers, Netlify, Render, Git, Jira
Systems & Networking: Linux, TCP/IP, HTTPS, DNS, SSH, MQTT, BLE, LTE, Wi-Fi
Spoken Languages: English (C1), French (B1)

Leadership & Community

- Served as IEEE AP-S Chapter Webmaster and Treasurer, maintaining chapter web content through **WordPress**.
- Served as Media Lead and Director of Service, mentoring youth, supporting academic instruction for high-school students in orphanages, and participating in weekly community outreach.